

Stripe

You are a Data Analyst on the Stripe Capital team, focused on assessing lending patterns for small businesses....

Question 1: For each business, what is the total transaction amount in January 2024, and how does its ranking compare to other businesses? This analysis will help establish baseline revenue activity levels for financing eligibility.

```
SELECT
  b.business_name,
  t.total_january_amount,
  DENSE_RANK() OVER (ORDER BY t.total_january_amount DESC) AS
ranking
FROM (
  SELECT
    business_id,
    SUM(transaction_amount) AS total_january_amount
  FROM fct_transactions
  WHERE transaction_date >= '2024-01-01'
    AND transaction_date < '2024-02-01'
  GROUP BY business_id
) AS t
INNER JOIN dim_businesses AS b
  ON t.business_id = b.business_id;
```



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Question 2: For each business, calculate the percentage change in total transaction amount from January 2024 to February 2024. This metric will assist in understanding recent revenue growth trends.

```
with january AS(
  SELECT business_id, SUM(transaction_amount) AS jan_total
  FROM fct_transactions
  WHERE transaction_date >= '2024-01-01'
    AND transaction_date < '2024-02-01'
  GROUP BY business_id),

  february AS(
  SELECT business_id, SUM(transaction_amount) AS feb_total
  FROM fct_transactions
  WHERE transaction_date >= '2024-02-01'
    AND transaction_date < '2024-03-01'
  GROUP BY business_id)

SELECT d.business_name, jan_total, feb_total,
  CASE
  WHEN jan_total = 0 THEN NULL
  ELSE (feb_total - jan_total) / jan_total * 100
  END AS pct_change
FROM january
INNER JOIN february USING(business_id)
INNER JOIN dim_businesses AS d USING(business_id)
```



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Question 3: For each business, compute the month-over-month growth in total transaction amounts from January 2024 through March 2024. Rank each business based on this average growth (from highest to lowest), and return the rank number along with the business name and average growth.

This analysis will inform which businesses exhibit stable and growing revenue patterns relevant to financing eligibility.

```
WITH monthly_totals AS (  
  SELECT  
    business_id,  
    TO_CHAR(transaction_date, 'YYYY-MM') AS month,  
    SUM(transaction_amount) AS total_amount  
  FROM fct_transactions  
  WHERE TO_CHAR(transaction_date, 'YYYY-MM') IN ('2024-01', '2024-02', '2024-03')  
  GROUP BY business_id, TO_CHAR(transaction_date, 'YYYY-MM'))  
  
calculate AS(  
  SELECT  
    d.business_id,  
    business_name,  
    month,  
    total_amount,  
    LAG(total_amount) OVER (PARTITION BY business_id ORDER BY month) AS previous_month_amount,  
    ROUND(  
      CASE  
        WHEN LAG(total_amount) OVER (PARTITION BY business_id ORDER BY month) IS NULL THEN NULL  
        WHEN LAG(total_amount) OVER (PARTITION BY business_id ORDER BY month) = 0 THEN NULL  
        ELSE ((total_amount - LAG(total_amount) OVER (PARTITION BY business_id ORDER BY month))  
              / LAG(total_amount) OVER (PARTITION BY business_id ORDER BY month)) * 100  
      END, 2) AS mom_growth_percentage  
  FROM monthly_totals  
  INNER JOIN dim_businesses AS d  
  USING (business_id)  
  ORDER BY business_id, month)  
  
SELECT business_name,  
  average_growth,  
  RANK() OVER(ORDER BY average_growth DESC) AS ranking  
FROM(  
  SELECT business_name,  
    AVG(mom_growth_percentage) AS average_growth  
  FROM calculate  
  GROUP BY business_name) AS final
```

